

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Artificial Intelligence**

Code No: **CSC266** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What are the problems of depth limited search? Use the Alpha-Beta pruning algorithm to prune the following game tree assuming child nodes are visited from left to right. Show all final alpha and beta values computed at root, each internal node explored, and at the top of pruned branches.
2. How do you infer knowledge from semantic net? Illustrate with an example. Consider the following Bayesian network with random variables A, B, C and D. If A is true, find the probability of D being true.
3. What is model free reinforcement learning? Differentiate between active and passive reinforcement learning. Describe the different operators used in genetic algorithm.

Section B

Attempt any Eight questions[8×5=40]

4. How do you define rational? Can AI choose between right and wrong? Justify.
5. What do you mean by PEAS description? Give an example.
6. Apply the hill climbing search algorithm to solve the following problem. Assume the heuristic function $h(n) = +1$ for all blocks in the support structure if the block is correctly positioned otherwise - 1.[5]Initial State ? A D C BGoal State ? D C B A
7. Why do we need posterior probability? Design a semantic network for following statements.Dogs hate cats. Tom is a cat. Puppy is a dog. Cats chases rats. Rats are clever.
8. Design a Hebb net to implement logical OR function.
9. Given the premises, all living things are either animal or plant. Plant needs sunlight for photosynthesis. Mustard is living thing but it is not an animal. Does mustard need sunlight for photosynthesis.
10. What are the phases for developing expert system? Explain.
11. Why do machine need to understand and generate natural language? What might be the challenges? Explain.

12. List any one examples of static and dynamic environment for agent. Why and when do you prefer rule based systems?